

Thesis progress report

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

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One loop, seven companion models

The rewrite was shown on a toy graph. The same compile commands are used here on the networks a companion computer would run.

There is no second compiler for YOLO. Each network is one record: a task, an ONNX file, and an export script. The pipeline never branches on the model name.

- yolov8n and yolo11n detect people and vehicles.
- yolov8n-pose detects body keypoints.
- yolov8n-seg produces masks.
- ssdlite_mobilenetv3 is a lighter detector.
- mobilenetv3_small is a small classifier.
- midas_small estimates depth from one camera.

Ultralytics YOLO graphs use SiLU, which is Conv, then Sigmoid, then Mul. The CPU fusion pass only joins Conv and Relu, so those YOLO graphs do not get smaller. MobileNet and MiDaS do get smaller where Conv is followed by batch-norm or Relu. That difference is a result.

There is no fused CPU kernel for SiLU in this project, and none will be invented to make the node count fall. The list of models lives in experiments/zoo.yaml. Adding a model means a new record and an export script, not a new branch in the compiler.

The first design was a YOLO compiler. That would have made every new camera task a new program. The node counts then looked like a failed fusion: several YOLO graphs stayed the same size. The pass was fine. Those graphs do not contain the Conv-ReLU pair it knows how to join. Writing a pretend SiLU kernel would have made the table look better and the system dishonest. MobileNet and MiDaS shrinking, on the same pass, is what showed the pass was alive.

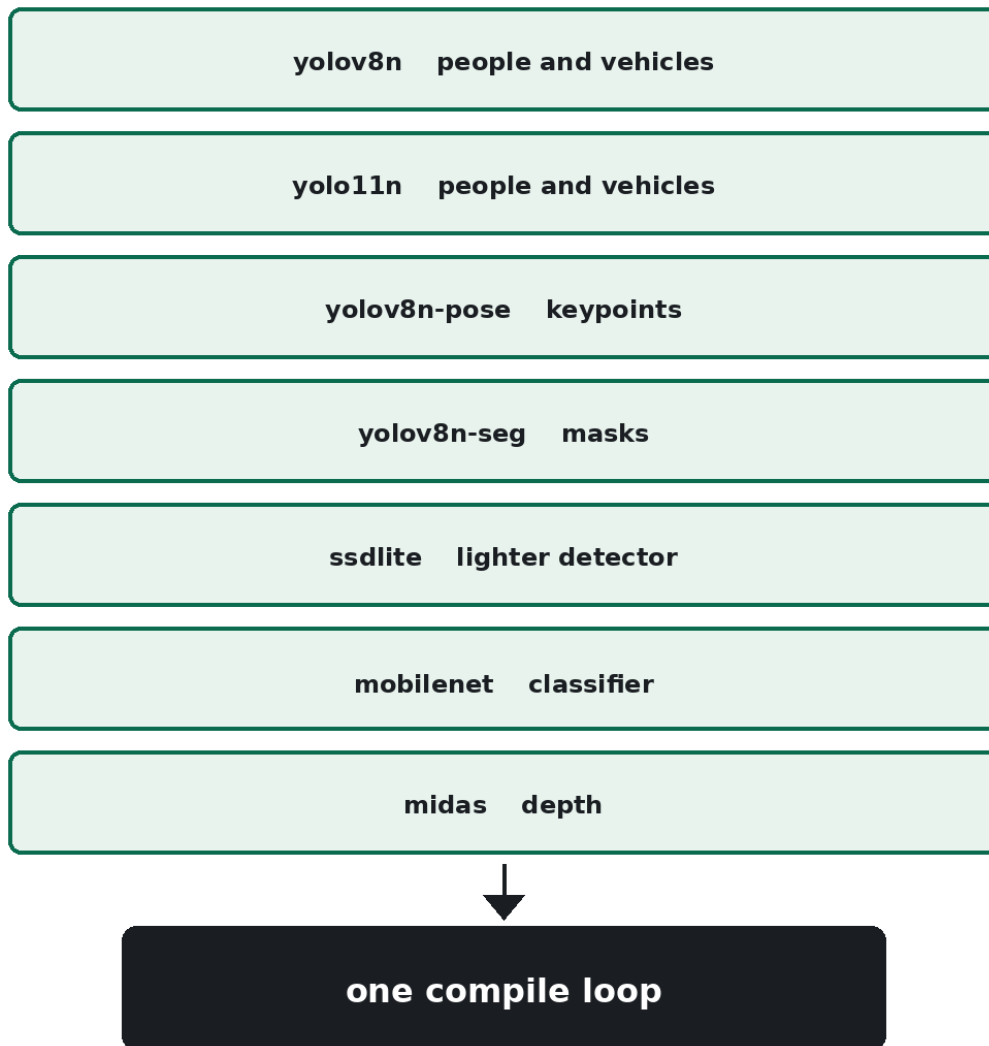


Figure 1. Seven models enter one compile loop.

One loop covers these seven models. A plan can still be refused before that loop rewrites anything.